

HW-4: AGC Design for 16-QAM

1. Design a MATLAB program to realize a simple transceiver, as shown in Fig. 1, including a mapper, a channel, a de-mapper, a slicer, and an AGC for 16-QAM modulation schemes, where the channel model is defined as $h_n = 0.5\delta_n$. Besides, both D/A and A/D can be viewed as ideal. A test bit streams (“bin.dat”) can be founded in E-Learning.

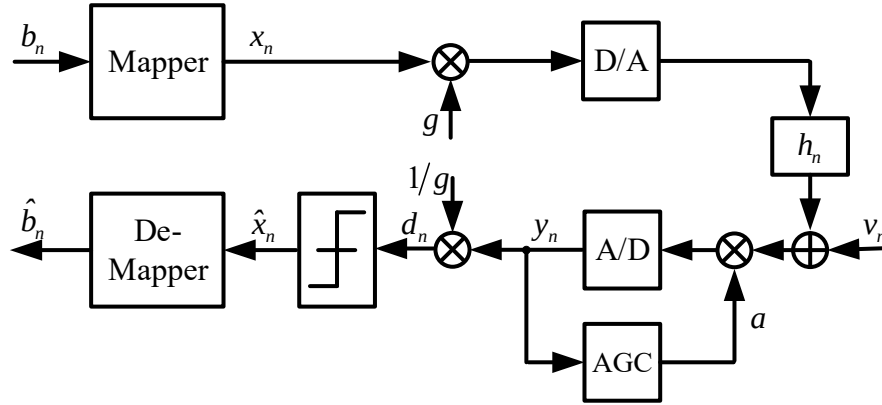


Fig. 1: Simple transceiver including a mapper, a channel, a de-mapper, a slicer, and an AGC.

Actually, the slicer is non-linear device that is applied to realize a series of decision thresholds for the input signal. The related operation of the slicer for 16-QAM, the gain scaling, and the white Gaussian noise are described as follows,

- (1) The decision thresholds of the slicer are 0 and ± 2 that can be easily found in Fig. 2. Considering an example for the real-part of an input signal, the slicer operation is described as

$$\hat{x}_{R,n} = \begin{cases} +3, & y_{R,n} \geq 2 \\ +1, & 0 \leq y_{R,n} < 2 \\ -1, & -2 \leq y_{R,n} < 0 \\ -3, & y_{R,n} < -2 \end{cases}$$

where $\hat{x}_{R,n}$ and $y_{R,n}$ are the real-part for $\hat{x}_n$ and y_n , respectively. Similarly, the imaginary-part of an input signal is the same as the decision thresholds of the real-part of an input signal.

- (2) The gain factors, g and $1/g$, are used to normalize the signal power of x_n to one, where g is defined as follows,

$$g = 1/\sqrt{10}$$

- (3) For 16-QAM, the white Gaussian noise with zero mean and unit standard deviation ($\sigma_v \approx 1$) can be modelled as follows,

- (a) The $v_n = v_{R,n} + jv_{I,n}$ is a complex value of the white Gaussian noise,

where $v_{R,n}$ and $v_{I,n}$ are the real and the imaginary part of v_n . A template MATLAB for white Gaussian noise is given as

$$v_n = \sigma_v * \left(\frac{\text{randn}(1)}{\sqrt{2}} + j * \frac{\text{randn}(1)}{\sqrt{2}} \right);$$

(4) As a matter of fact, $\hat{x}_n = x_n$ and $\hat{b}_n = b_n$ for 16-QAM

Problem:

Q1. Generate -25 dB of white Gaussian noise for 16-QAM.

Q3. Please design an AGC algorithm using an amplitude detector and a moving sum to realize AGC operation and to make the subsequent functions work well. In other words, once the AGC is convergence, the received data can be completely detected by slicer. Therefore, you only need to check the last 100 symbols.

According to the block diagram of AGC as illustrated in class, the amplitude of a complex signal $x_n = x_{R,n} + jx_{I,n}$ can be expressed by $A_n = \sqrt{x_{R,n}^2 + x_{I,n}^2}$. Besides, the moving sum can be fulfilled as $\frac{1}{N} \sum_{n=0}^{N-1} A_n$, where N can be 4, 8, or 16. The V_{ref} is equivalent to the average amplitude of 16-QAM.

Q3. You have to deliver the following two things to E-Learning.

- (1) .m file
- (2) AGC algorithm including math derivation and description for your AGC algorithm